Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-08-27 07:30:02

Situation Summary

Stage IV: 5 of 33
Biomarker Count: 21 entries
Top Targets: ctDNA (7) · MSI/MMR (5) · NGS (3)
Breakthrough: 11 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED AUG 24, 2026 9

Liquid biopsy in colorectal cancer: redefining surgical decision-making

Liquid biopsy (LB) provides a minimally invasive method for detecting circulating tumor DNA (ctDNA) in colorectal cancer, offering insights into tumor biology that enhance postoperative management. This scoping review highlights LB's potential to inform surgical decision-making and improve risk stratification beyond traditional anatomical assessments.

pubmed.ncbi.nlm.nih.gov

HIGH Stage II ctDNA VERIFIED LIQUID BIOPSY PUBLISHED AUG 09, 2026 6

Circulating tumor DNA in gastrointestinal cancers: promise, pitfalls, and the path forward

Postoperative circulating tumor DNA (ctDNA) positivity in colorectal cancer identifies patients at high risk of recurrence, with serial clearance patterns enhancing risk stratification. However, the prognostic validity of ctDNA has not consistently translated into treatment-selection utility. The DYNAMIC trial demonstrated that ctDNA-guided chemotherapy de-escalation in stage II colon cancer does not compromise recurrence-free survival.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED AUG 21, 2026 6

Beyond tissue biopsy: ctDNA methylation analysis for prediction of colorectal cancer prognosis

ctDNA methylation analysis offers a novel approach for prognostic stratification in colorectal cancer, addressing limitations of mutation-based methods that focus solely on genetic alterations. This study highlights the potential of ctDNA methylation to provide deeper insights into tumor biology and improve postoperative risk assessment by better capturing molecular residual disease (MRD).

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED AUG 19, 2026 5

Early-Onset Colorectal Cancer With Liver-Only Metastases: A Retrospective Cohort Study Integrating Prospectively Collected Real-World Clinical and Molecular Data From an Australian National Database (2009-2024) to Guide Treatment Planning

Early-onset colorectal cancer (EOCRC) exhibits distinct biological differences compared to liver-only metastatic colorectal cancer (LOCRC), including variations in sex distribution and BRAF mutation rates. In a retrospective cohort study utilizing real-world clinical and molecular data, liver resection was linked to improved survival in LOCRC. Treatment benefits varied based on age, primary tumor site, and the timing of liver metastases, indicating the need for tailored treatment planning.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage IV NGS VERIFIED GENOMIC PROFILING PUBLISHED AUG 26, 2026 4

Integrated NGS and ddPCR analysis of RAS/RAF mutations in metastatic colorectal cancer among Kazakhstani patients: a pilot study

KRAS mutations are the predominant genetic alterations in metastatic colorectal cancer, with BRAF mutations being less frequent. The study utilized targeted next-generation sequencing and droplet digital PCR to enhance mutation detection accuracy in FFPE samples. These findings elucidate the molecular characteristics of metastatic colorectal cancer in Kazakhstan and support the advancement of precision oncology strategies in clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED AUG 07, 2026 4

Genomic Landscape and Therapeutic Implications of ALK Fusion-Positive Colorectal Cancer

ALK fusions represent a rare subset of colorectal cancer (CRC) with a distinct genomic profile, often associated with MSI-high/MMR-deficient status. Comprehensive genomic profiling is recommended to identify ALK fusions, which may guide the use of ALK-directed therapies in this population. The study underscores the frequent co-occurrence of MSI-high disease in ALK fusion-positive CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages NGS VERIFIED GENOMIC PROFILING PUBLISHED AUG 13, 2026 3

Next-Generation Sequencing in Colorectal Cancer: Real-World Molecular Profiling and Clinical Correlations

Next-generation sequencing (NGS) was utilized in a retrospective single-centre study to characterize the molecular landscape of 97 clinically selected colorectal cancer patients. The study explored associations between genomic alterations identified through targeted NGS and clinicopathological features, highlighting the potential for improved patient stratification in routine clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED JUL 30, 2026 3

Independent and sex-stratified association between microsatellite instability and peripheral hemoglobin in colorectal cancer

Lower peripheral hemoglobin (HB) levels are independently associated with microsatellite instability-high (MSI-H) status in colorectal cancer, though the discrimination is modest. While HB may serve as an adjunctive signal, it is not a substitute for tumor-based MMR/MSI testing. Further prospective external validation is necessary.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BRAF VERIFIED GENOMIC PROFILING PUBLISHED JUL 30, 2026 3

Performance Evaluation of a Colorectal Cancer-Associated MassARRAY-Based Hotspot Genotyping Assay for KRAS, NRAS, BRAF, and PIK3CA Using Solid Tumor Specimens

The iPLEX® HS Colon Panel demonstrates reliable performance for rapid detection of clinically actionable hotspot mutations in KRAS, NRAS, BRAF, and PIK3CA using mixed FFPE tumor specimens. Analytical validation supports its use, with further evaluation in larger colorectal cancer cohorts recommended.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage II KRAS VERIFIED RESEARCH PUBLISHED AUG 27, 2026 2

KRAS Mutation Detection by Real-Time and Digital PCR in Tumor Tissue and Plasma and Its Association with Survival in Stage II-IV Colorectal Cancer: A Kazakhstan Cohort Study

KRAS mutation detection in tumor tissue and plasma was assessed using real-time and digital PCR in a Kazakhstan cohort of Stage II-IV colorectal cancer. The study found that variant-level reporting provides more prognostic information than binary calls. Additionally, a higher detection rate in plasma using digital PCR was observed, suggesting the need for further prospective studies with standardized mutation panels.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages TMB VERIFIED RESEARCH PUBLISHED AUG 13, 2026 2

Co-Mutation of CREBBP/EP300 and POLE/POLD1 Identifies a TMB-High Subset of MSS CRC

Co-mutation of CREBBP/EP300 and POLE/POLD1 identifies a subset of TMB-high microsatellite stable (MSS) colorectal cancer (CRC). Analysis of multiple public cohorts demonstrated that CREBBP/EP300 mutations are independently associated with TMB-high status ($p < 0.00001$). This finding suggests potential for these mutations as biomarkers for immunotherapy eligibility in MSS CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED AUG 13, 2026 2

Factors Associated with the Timing of Liver Metastasis After Colorectal Cancer Surgery: A Retrospective Multicenter Study

KRAS and BRAF^(V600E) mutations, along with various clinicopathological factors, were assessed for their association with the timing of liver metastasis in colorectal cancer patients post-surgery. The study aimed to establish a temporal threshold to differentiate between "early" and "late" metastasis. Findings suggest that specific mutations may influence the timing of liver metastasis, which could inform surveillance strategies post-surgery.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED GENOMIC PROFILING PUBLISHED JUL 28, 2026 2

Unique characteristics of Lynch syndrome-associated mismatch repair-deficient colorectal intramucosal carcinoma

Lynch syndrome-associated mismatch repair-deficient (MMR-D) intramucosal carcinomas (IMCs) exhibit distinct clinicopathologic and immunophenotypic profiles compared to MMR-proficient (MMR-P) and sporadic MMR-D IMCs. These findings underscore the necessity for tailored diagnostic and management strategies for patients with Lynch syndrome.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection BROAD SCOPE VERIFIED EARLY DETECTION PUBLISHED AUG 13, 2026 1

Genetic Landscape of Lynch Syndrome in a High-Risk Serbian Cohort: Predominance of *MLH1* Variants and Implications for Risk-Based Testing

In a high-risk Serbian cohort, the study identifies a predominance of pathogenic or likely pathogenic variants in the MLH1 gene associated with Lynch syndrome. The research highlights the need for risk-based testing in this population to improve hereditary colorectal cancer screening and management.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA
Dynamic Circulating Tumor DNA Methylation Monitoring Guiding Postoperative Surveillance in Nonmetastatic Colorectal Cancer: A Prospective, Randomized, Phase III FIND Trial

Dynamic monitoring of circulating tumor DNA (ctDNA) methylation was assessed in a prospective, randomized, Phase III FIND trial for postoperative surveillance in nonmetastatic colorectal cancer. The study indicates that ctDNA methylation can guide surveillance strategies, potentially improving early detection of recurrence.

MSI/MMR
EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

KRAS
Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C–mutated colorectal cancer (mCRC): Results from phase 3 CodeBreaK 300.

In the phase 3 CodeBreaK 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C–mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

TP53
Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF
Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS
Rapid On-Site Next-Generation Sequencing: An Alternative to Single-Gene and Send-Out Testing in Non-Small Cell Lung Cancer and Colorectal Cancer in a Community Pathology Laboratory Setting

Next-generation sequencing (NGS) using the ThermoFisher Oncomine Precision Assay (OPA) was assessed as an alternative to single-gene panels and send-out NGS for identifying actionable mutations in colorectal cancer. The study focused on turnaround time, quantity not sufficient rates, and detection of NCCN-recommended alterations, indicating that OPA may enhance efficiency in mutation identification for targeted therapies in a community pathology laboratory setting.

RAS
The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage, suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

Methylation

Lactate as a Master Regulator of Immune Suppression: From Metabolic Waste to Epigenetic Checkpoint in Colorectal Cancer

Lactate, a byproduct of aerobic glycolysis driven by KRAS and BRAF mutations in microsatellite-stable colorectal cancer, acts as a central immunosuppressive regulator in the tumor microenvironment. Its accumulation leads to immune suppression, contributing to resistance against immune checkpoint inhibitors. This highlights lactate's role beyond metabolic waste, suggesting potential therapeutic targets to enhance immune response in this cancer subtype.

TMB

Co-Mutation of CREBBP/EP300 and POLE/POLD1 Identifies a TMB-High Subset of MSS CRC

Co-mutation of CREBBP/EP300 and POLE/POLD1 identifies a subset of TMB-high microsatellite stable (MSS) colorectal cancer (CRC). Analysis of multiple public cohorts demonstrated that CREBBP/EP300 mutations are independently associated with TMB-high status ($p < 0.00001$). This finding suggests potential for these mutations as biomarkers for immunotherapy eligibility in MSS CRC.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **ctDNA, MSI/MMR, KRAS, TP53, and BRAF**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-08-27 07:30:02
This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

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Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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